

응력 해석 보고서



분석된 파일:	Prop Guard.iam
Autodesk Inventor 버전:	2024.4 (Build 284460000, 460)
작성 날짜:	2026-05-19, 오후 6:04
학습 작성자:	DELL
요약:	

정적 해석:1

일반 목표 및 설정:

설계 목표	단일 점
학습 유형	정적 해석
마지막 수정 날짜	2026-05-19, 오후 6:00
모형 상태	[1차]
설계 뷰	Default
위치	[1차]
강체 모드 탐지 및 제거	예
접촉 표면에서의 개별 응력	예
동작 하중 분석	아니오

iProperties

요약

작성자	Muhyidin
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프로젝트

부품 번호	Prop Guard
설계자	Muhyidin
추정 비용	₩0
작성 날짜	2020-09-01

상태

설계 상태	진행 중인 작업
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물리적 특성

질량	0.0863377 kg
면적	50785.6 mm^2
체적	81450.7 mm^3
무게 중심	x=-5.28277 mm y=33.2677 mm z=-81.3238 mm

참고: 물리적 특성 값이 아래 보고된 FEA에 사용된 값과 다를 수 있습니다.

메쉬 설정:

평균 요소 크기(모형 지름의 일부)	0.1
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최소 요소 크기(평균 크기의 일부)	0.2
등급 계수	1.5
최대 회전 각도	60 deg
곡선 메쉬 요소 작성	아니오
조립품 메쉬에 대한 부품 기반 측정 사용	예

재질

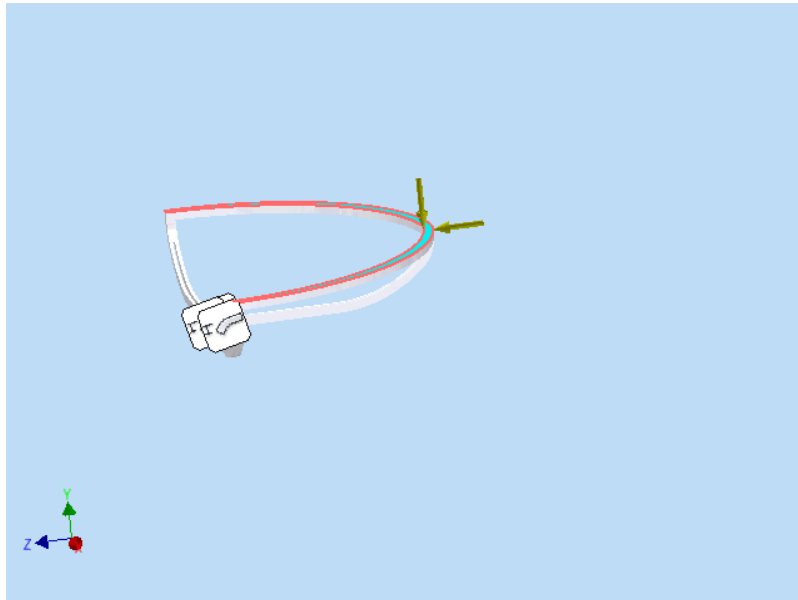
이름	ABS 플라스틱	
일반	질량 밀도	1.06 g/cm ³
	항복 강도	20 MPa
	극한 인장 강도	29.6 MPa
응력	탄성계수	2.24 GPa
	프와송의 비	0.38 ul
	전단 계수	0.811594 GPa
부품 이름	Prop Guard 2.ipt Prop Guard 1.ipt Prop Guard 3.ipt	

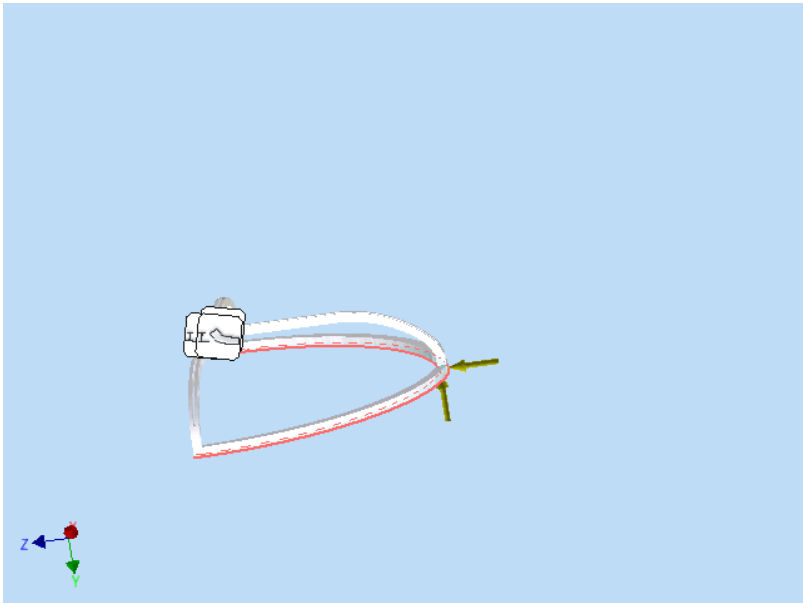
작업 조건

힘:1

하중 유형	힘
크기	500.000 N
벡터 X	0.000 N
벡터 Y	-500.000 N
벡터 Z	0.000 N

선택한 면

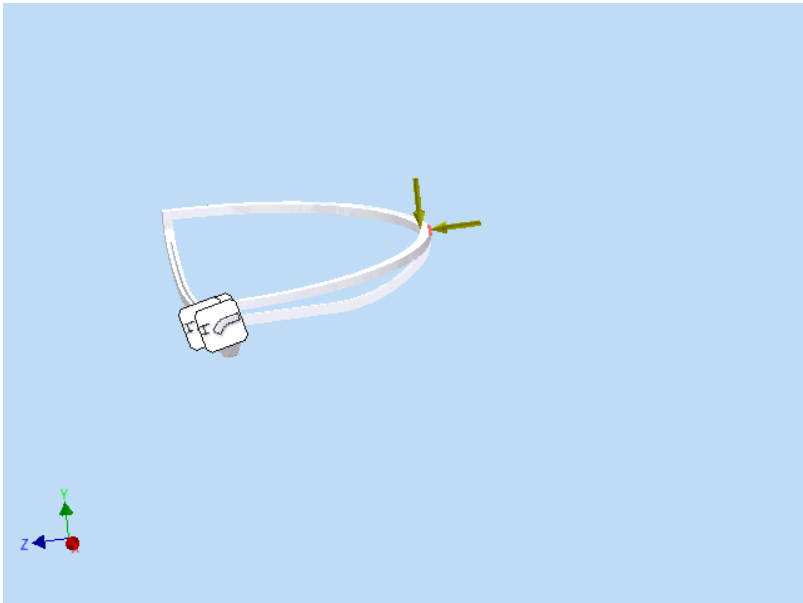


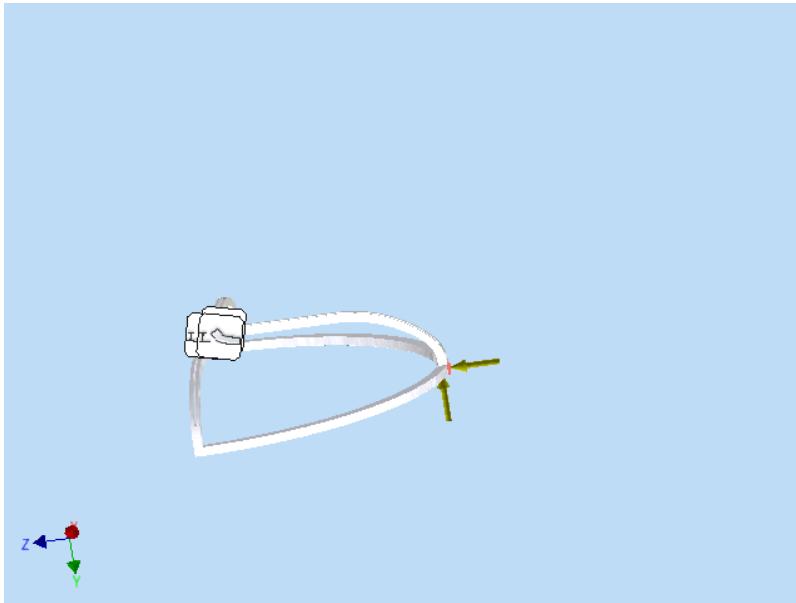


힘:2

하중 유형	힘
크기	500.000 N
벡터 X	0.000 N
벡터 Y	0.000 N
벡터 Z	500.000 N

선택한 면

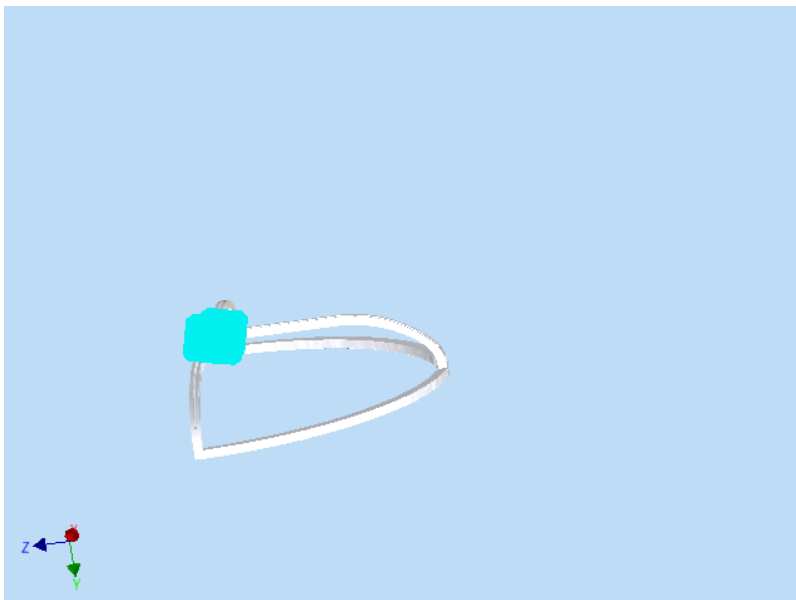
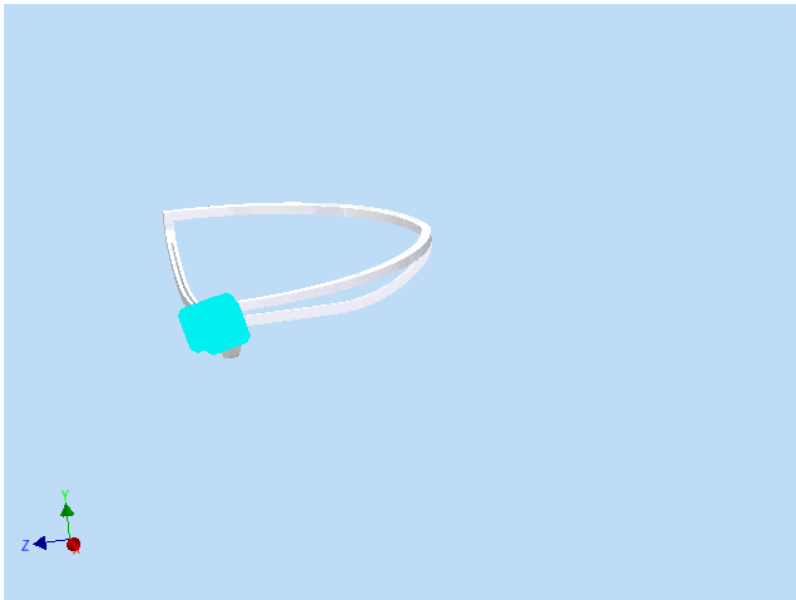




고정 구속조건:1

구속조건 유형 고정 구속조건

선택한 면



접촉 (접착)

이름	부품 이름
접착:1	Prop Guard 1:1 Prop Guard 2:3
접착:2	Prop Guard 1:1 Prop Guard 2:1
접착:3	Prop Guard 1:1 Prop Guard 2:2
접착:4	Prop Guard 1:1 Prop Guard 2:2
접착:5	Prop Guard 1:1 Prop Guard 2:1
접착:6	Prop Guard 2:1 Prop Guard 3:1
접착:7	Prop Guard 2:2 Prop Guard 3:1
접착:8	Prop Guard 2:3 Prop Guard 3:1

결과**구속조건에서의 반력 및 반작용 모멘트**

구속조건 이름	반력		반작용 모멘트	
	크기	구성요소(X,Y,Z)	크기	구성요소(X,Y,Z)
고정 구속조건:1	707.107 N	0 N	30.8459 N m	30.8459 N m
		500 N		0 N m
		-500 N		0 N m

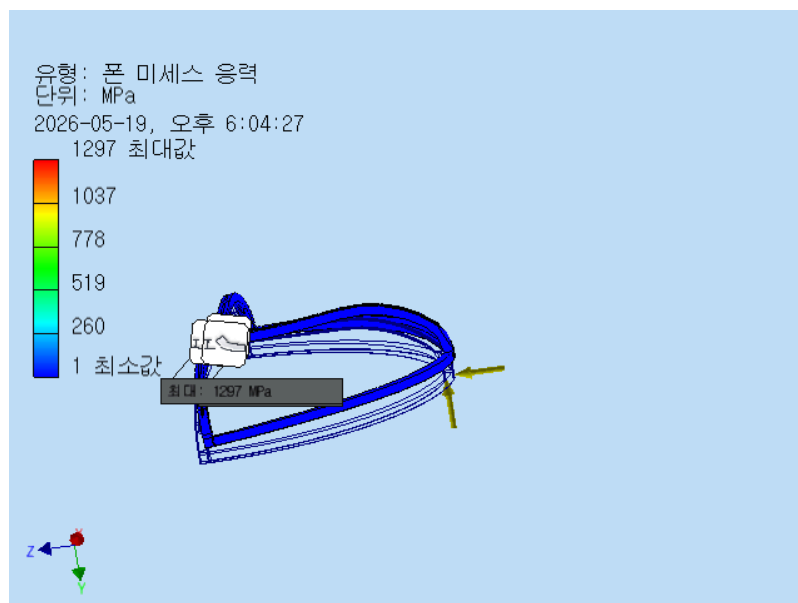
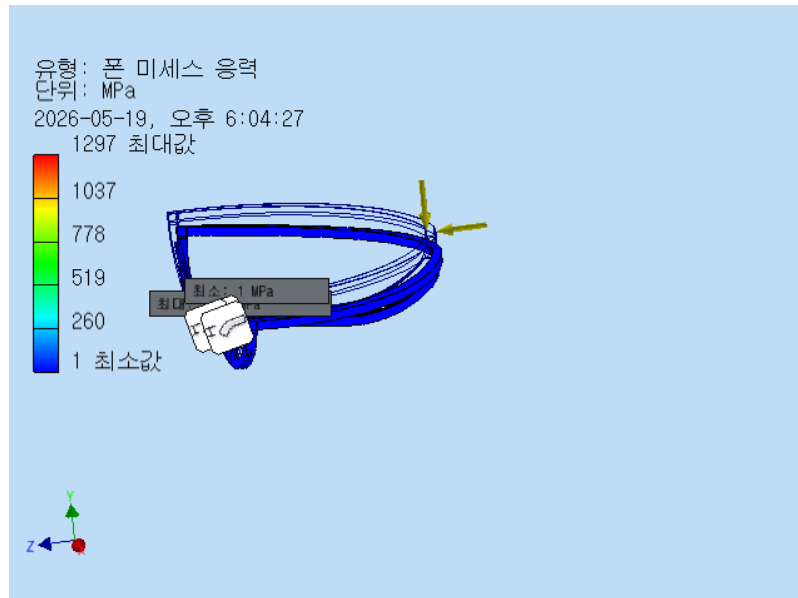
결과 요약

이름	최소값	최대값
체적	81450.7 mm ³	
질량	0.0863377 kg	
폰 미세스 응력	0.555456 MPa	1296.58 MPa
첫 번째 주 응력	-1035.88 MPa	2281.77 MPa
세 번째 주 응력	-2309.75 MPa	1190.7 MPa
변위	0 mm	368.45 mm
안전계수	0.0154252 ul	15 ul
응력 XX	-2055.76 MPa	2130.76 MPa
응력 XY	-447.818 MPa	449.797 MPa
응력 XZ	-374.13 MPa	386.931 MPa
응력 YY	-1181.39 MPa	1308.43 MPa
응력 YZ	-111.954 MPa	268.291 MPa
응력 ZZ	-1207.52 MPa	1301.88 MPa
X 변위	-104.779 mm	104.726 mm
Y 변위	-352.839 mm	0.550448 mm
Z 변위	-126.974 mm	8.69808 mm
등가 변형	0.000229357 ul	0.576594 ul
첫 번째 주 변형	-0.00151056 ul	0.601442 ul
세 번째 주 변형	-0.680401 ul	-0.000172106 ul
변형 XX	-0.596937 ul	0.547492 ul
변형 XY	-0.275888 ul	0.277107 ul
변형 XZ	-0.230491 ul	0.238377 ul
변형 YY	-0.107315 ul	0.121814 ul
변형 YZ	-0.0689718 ul	0.165287 ul
변형 ZZ	-0.265715 ul	0.197909 ul

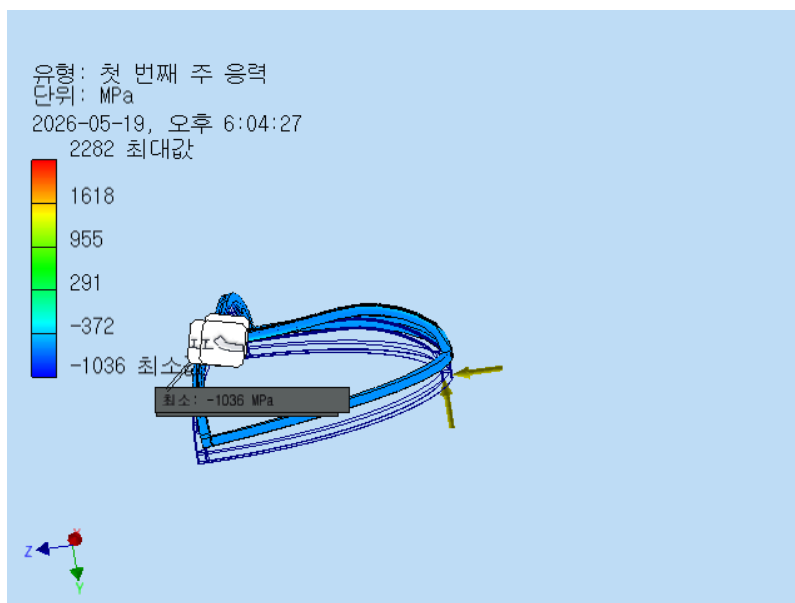
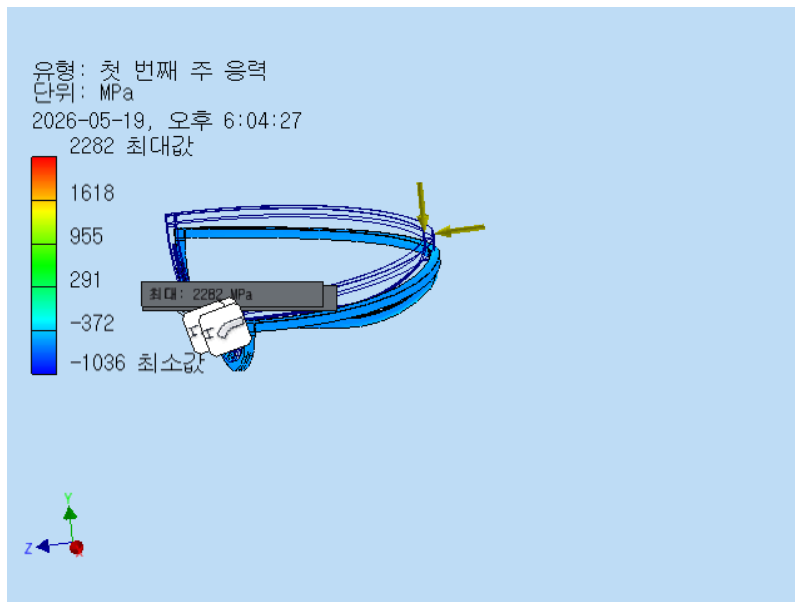
접촉 압력	0 MPa	1924.32 MPa
접촉 압력 X	-1527.76 MPa	1845.17 MPa
접촉 압력 Y	-487.462 MPa	202.546 MPa
접촉 압력 Z	-671.273 MPa	620.231 MPa

그림

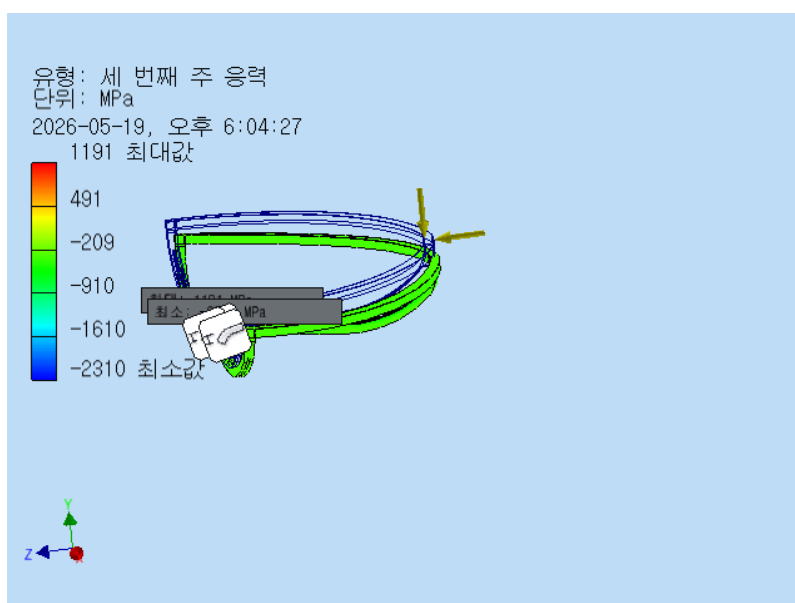
폰 미세스 응력

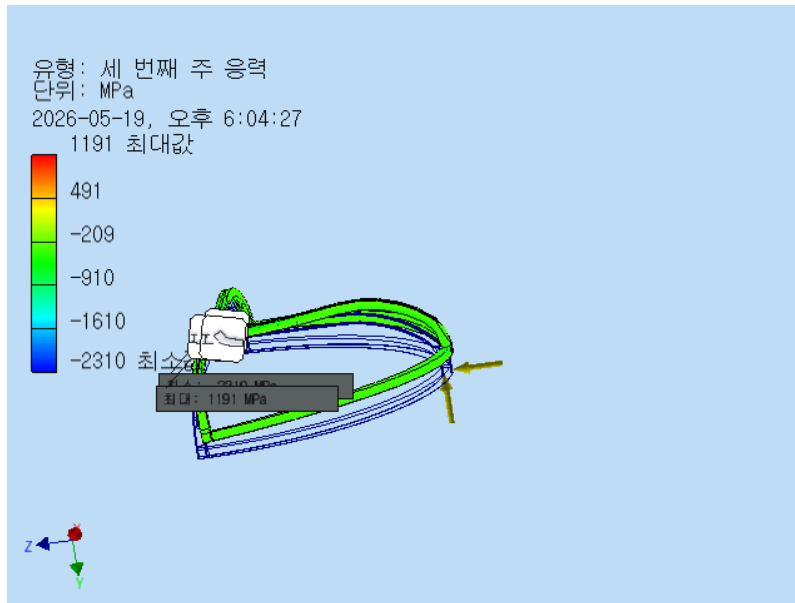


첫 번째 주 응력

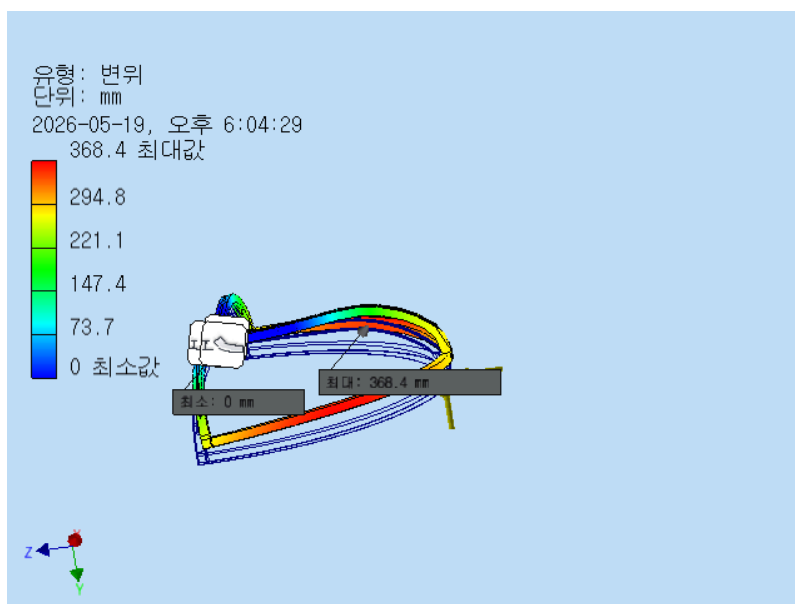
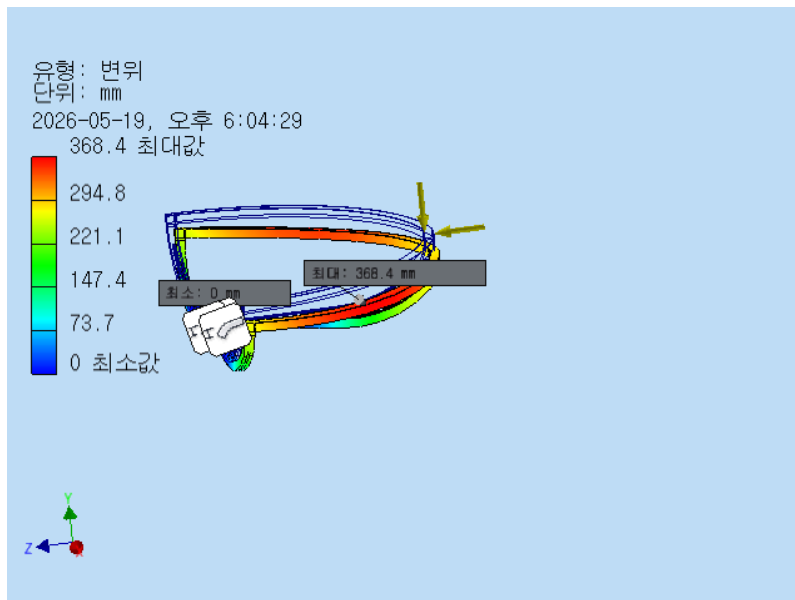


세 번째 주 응력

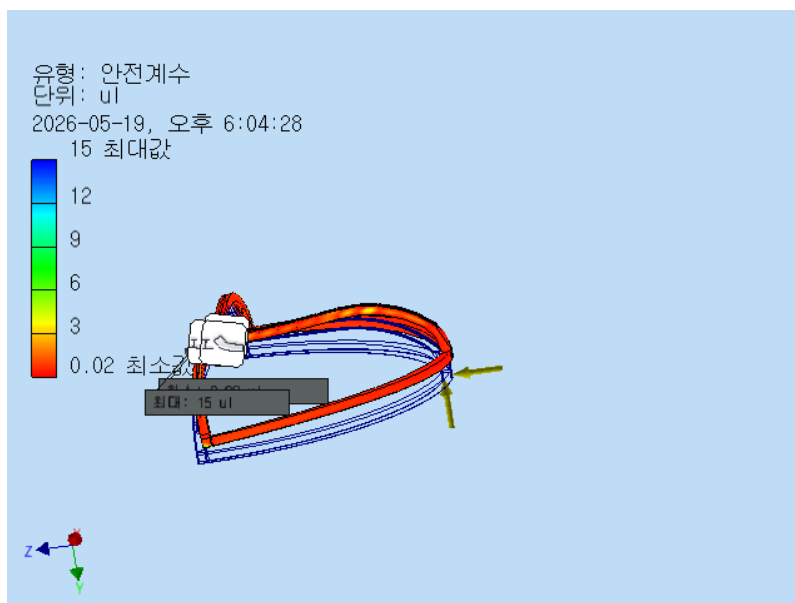
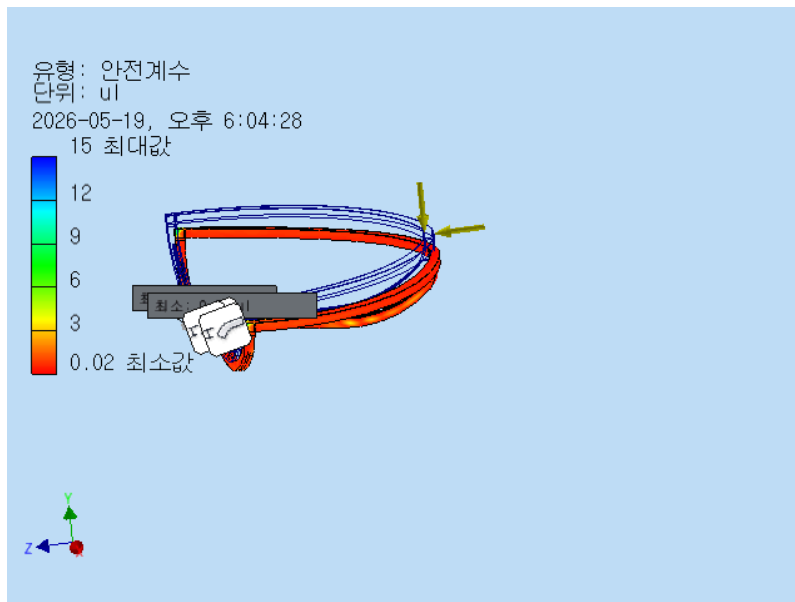




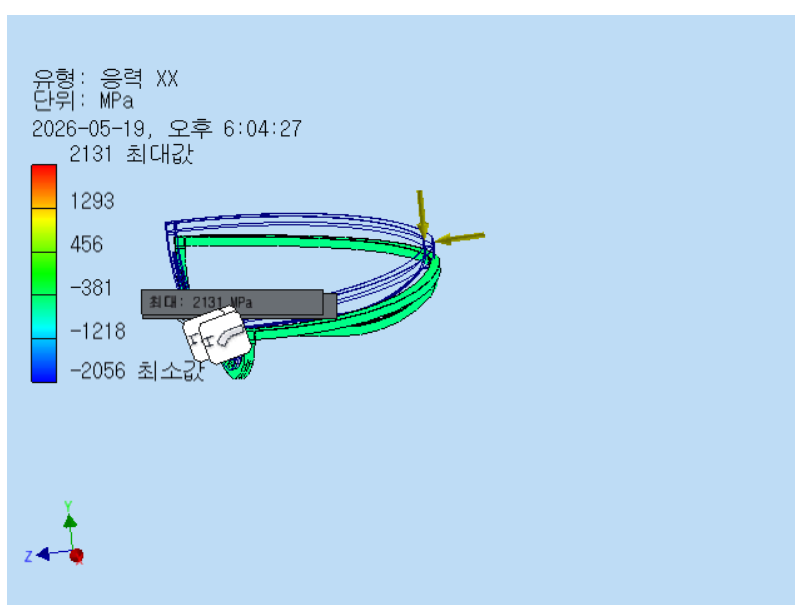
변위

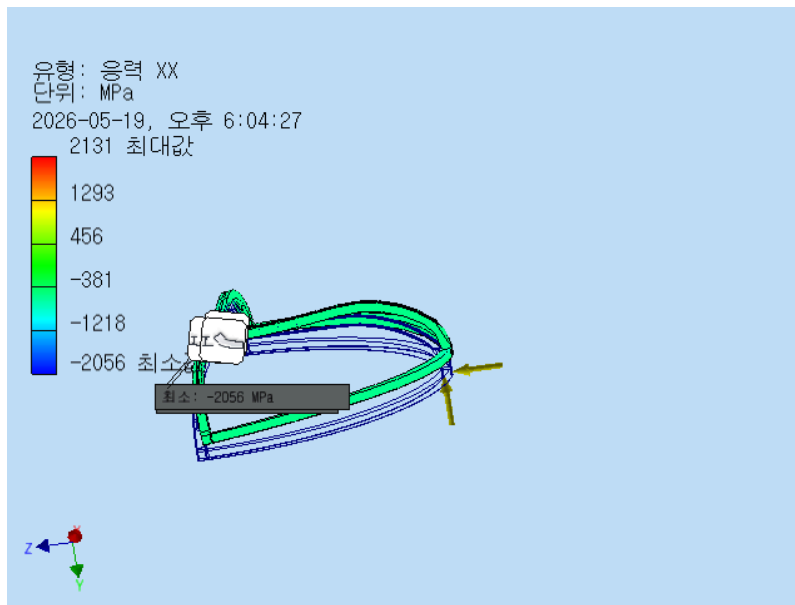


안전계수

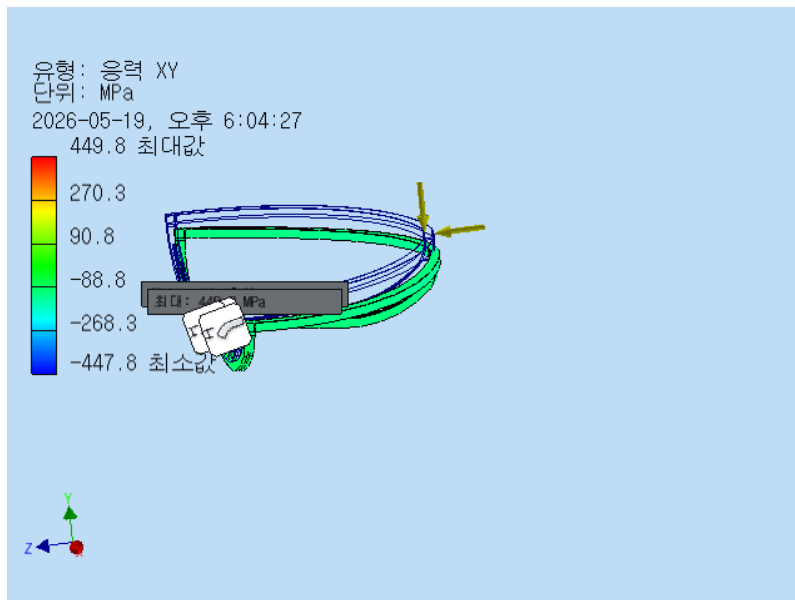


응력 XX

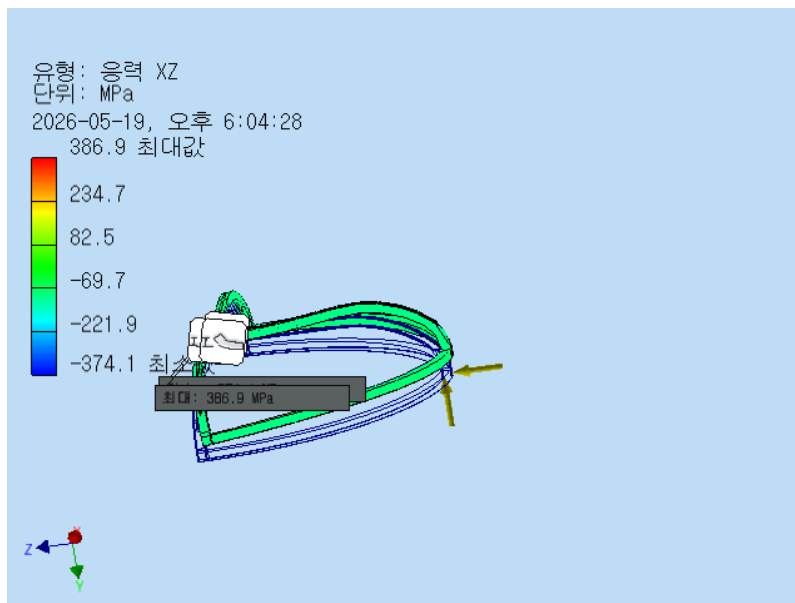
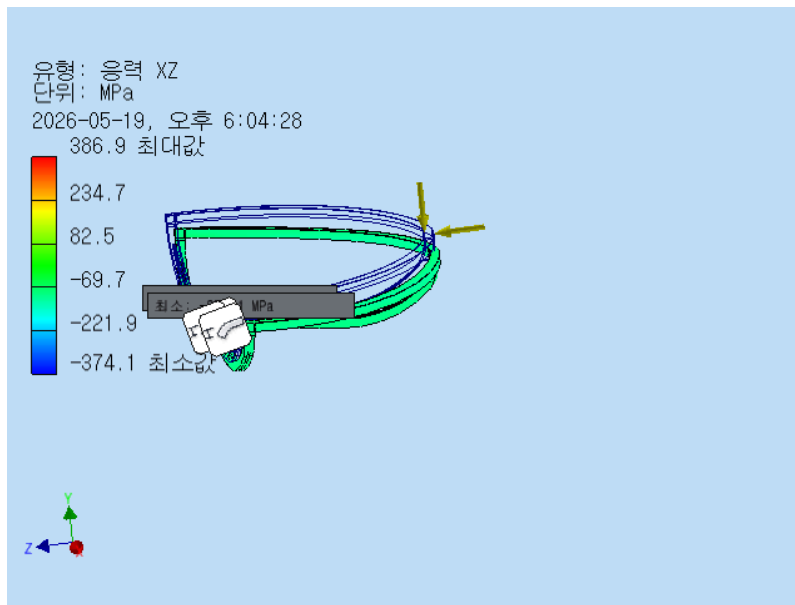




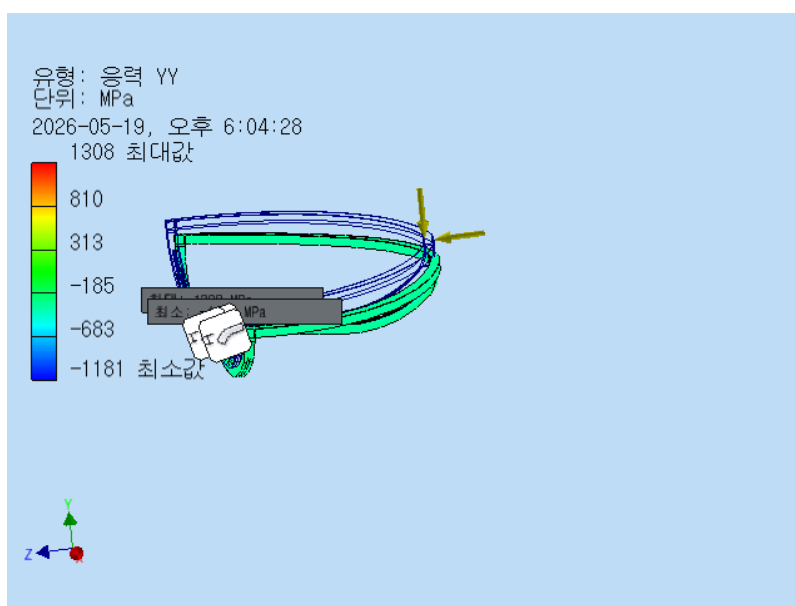
응력 XY

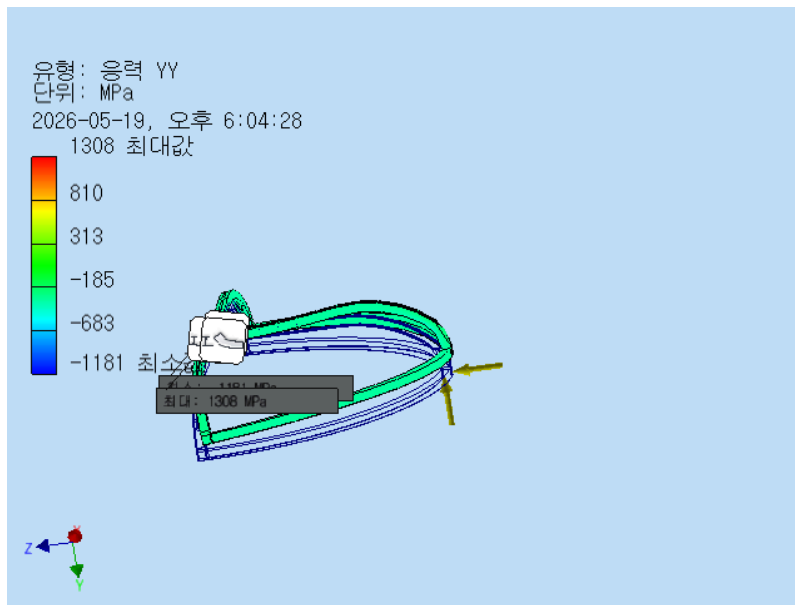


응력 XZ

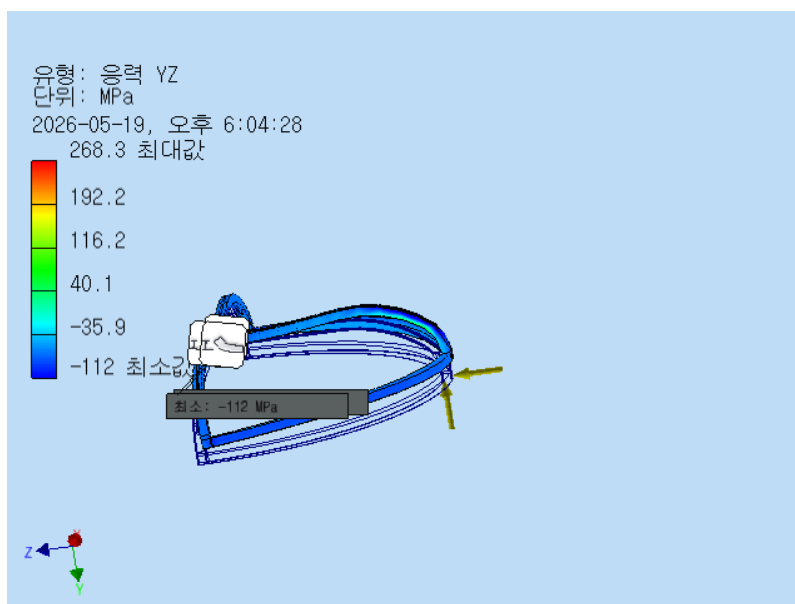
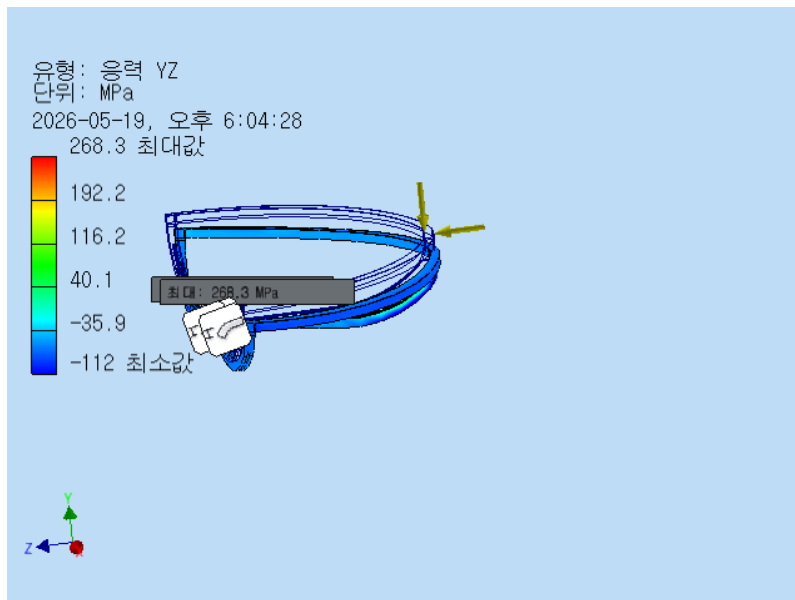


응력 YY

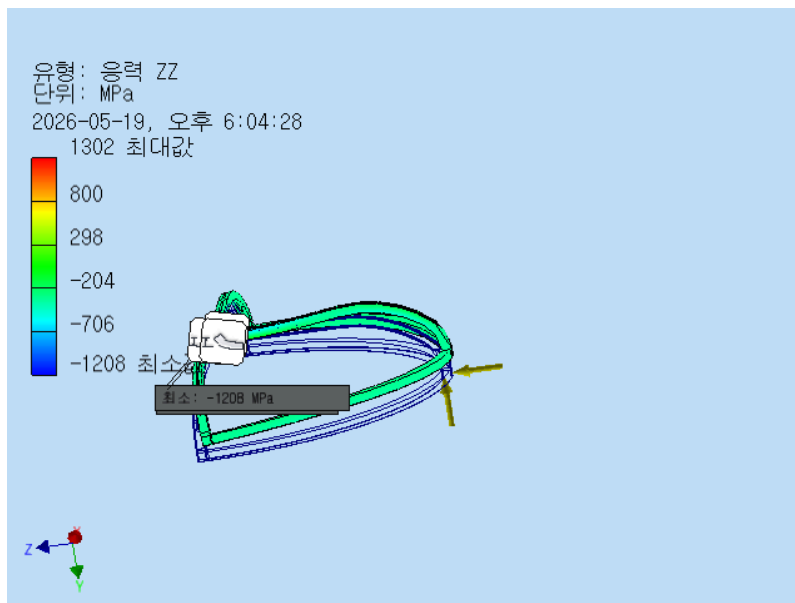
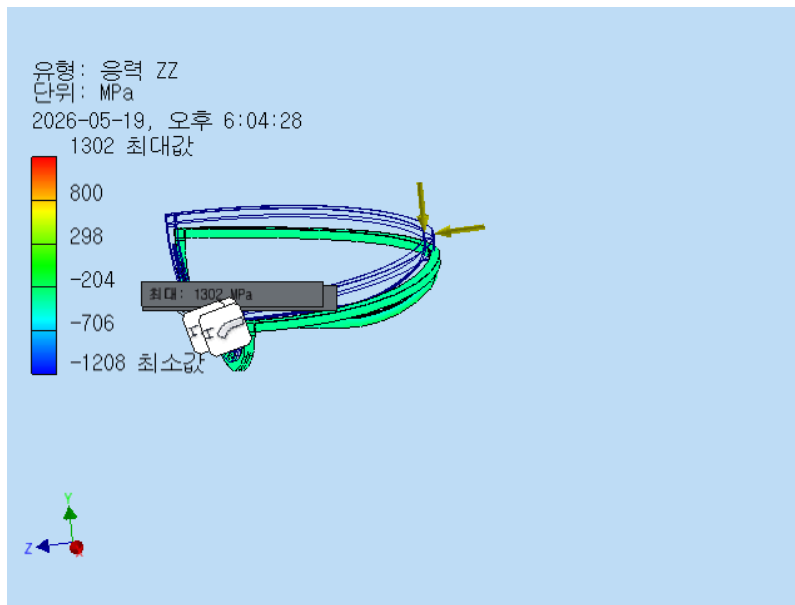




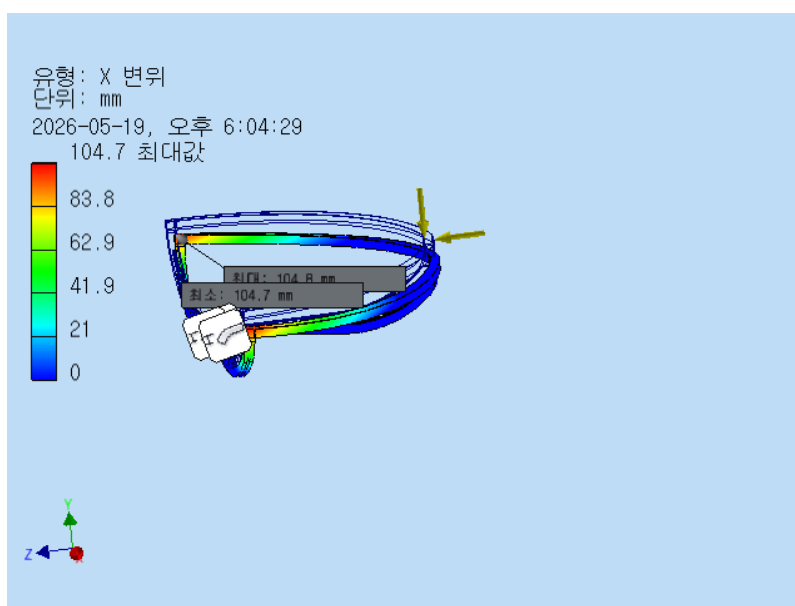
응력 YZ

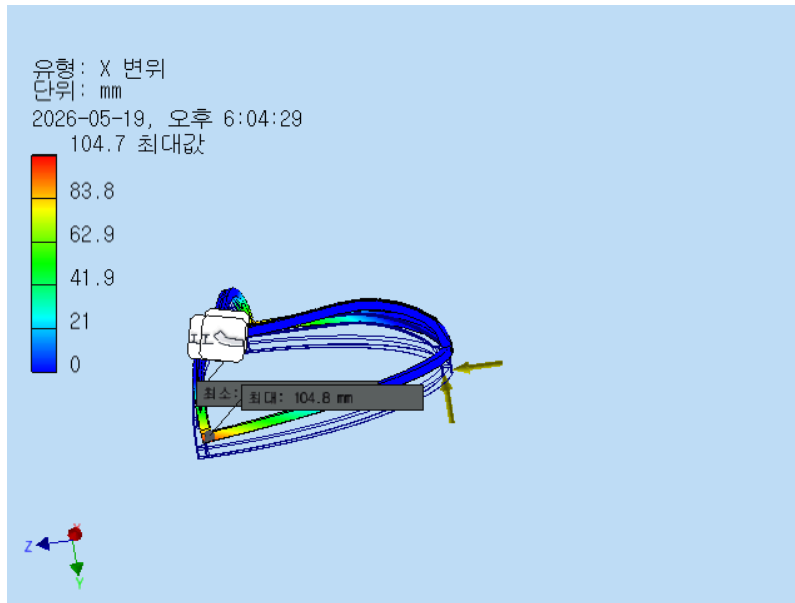


응력 ZZ

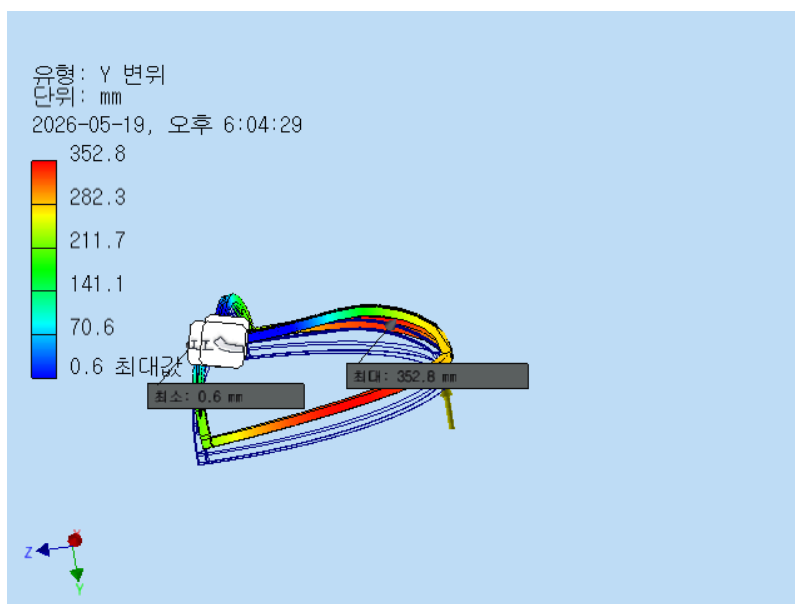
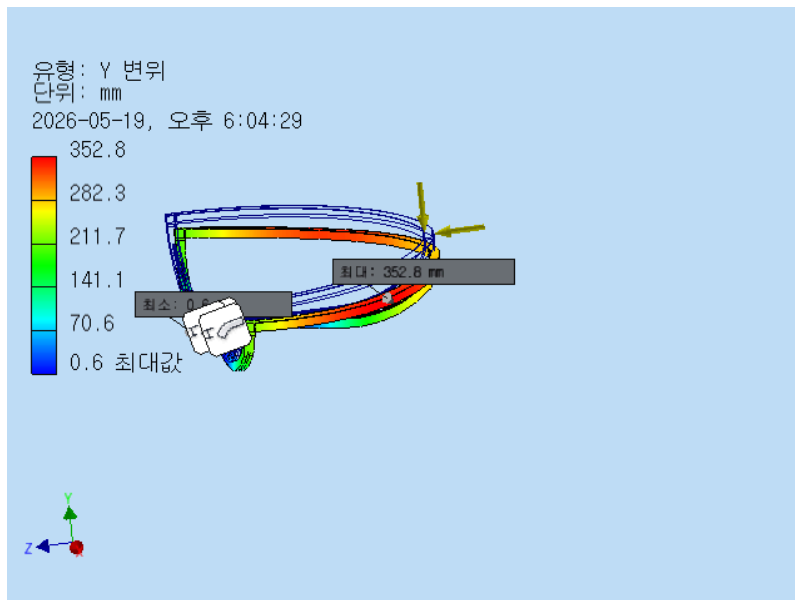


X 변위

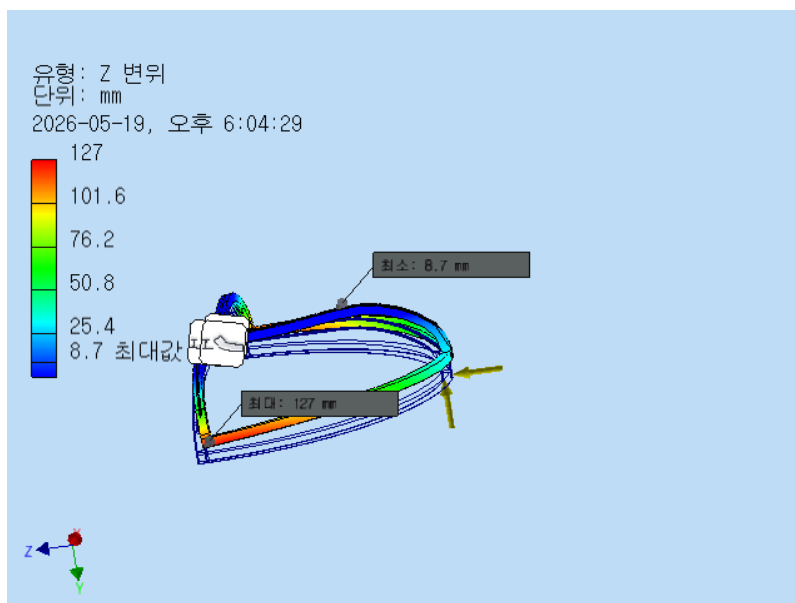
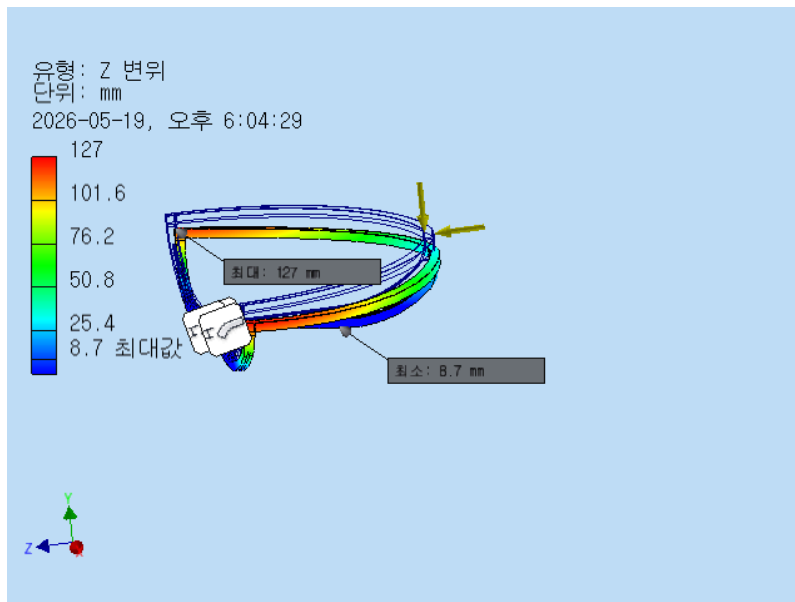




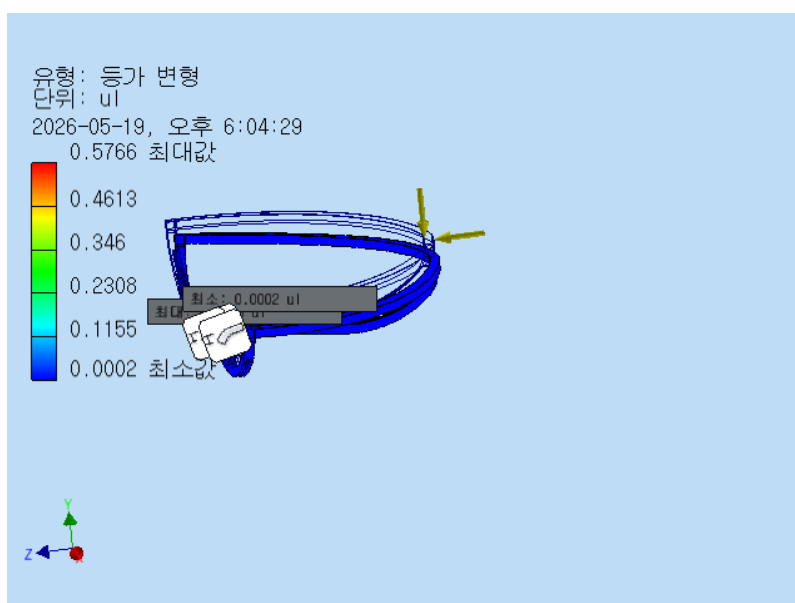
Y 변위

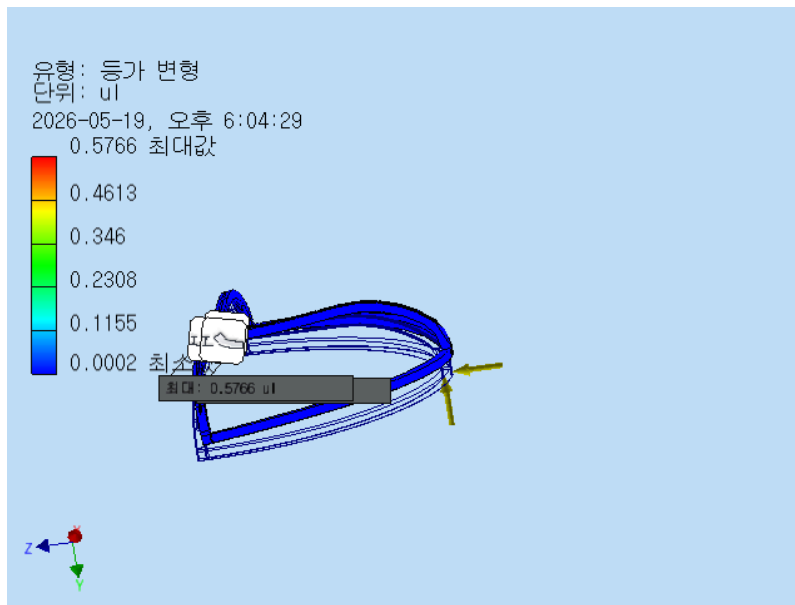


Z 변위

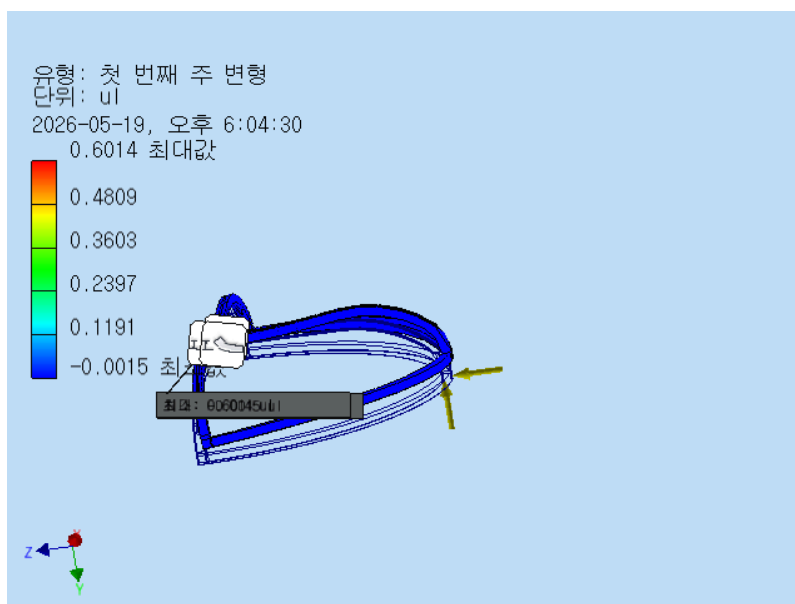
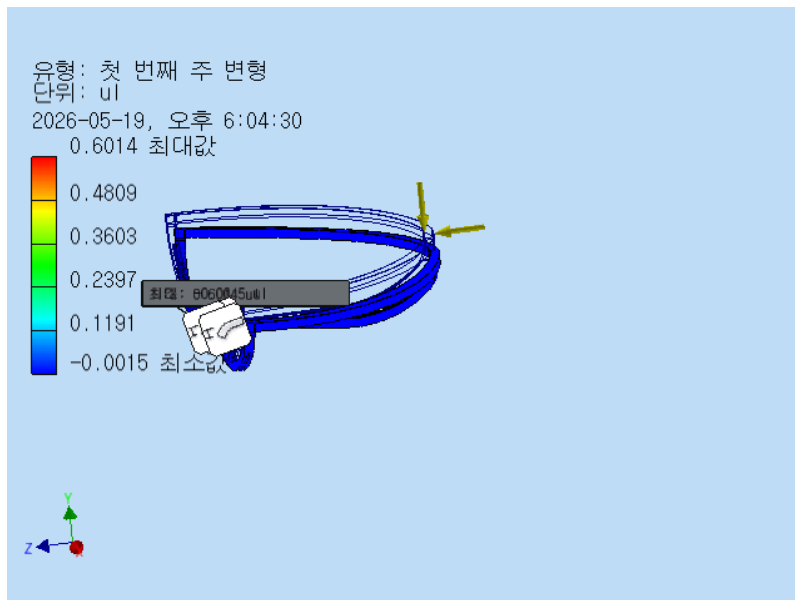


등가 변형

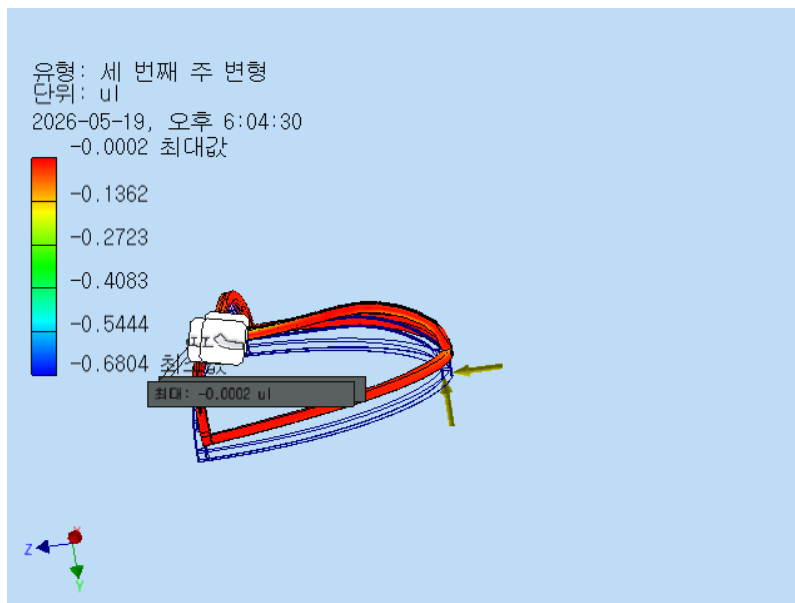
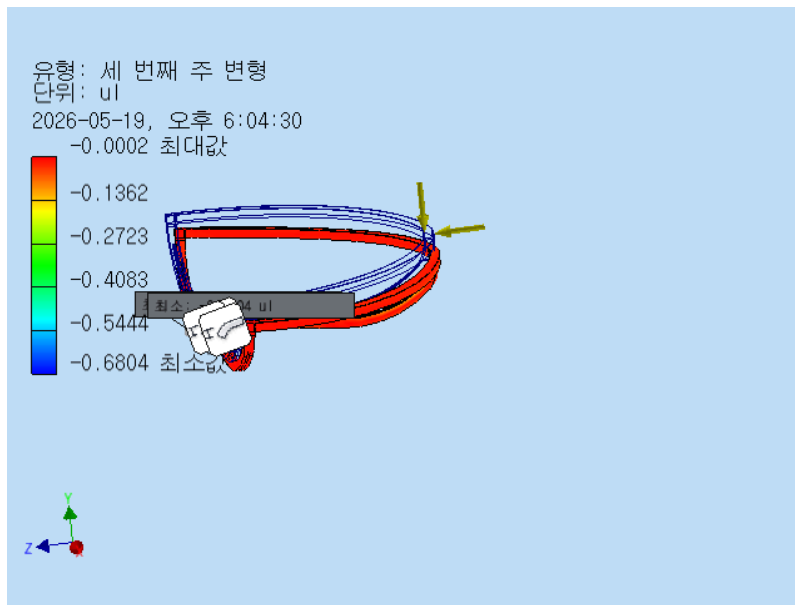




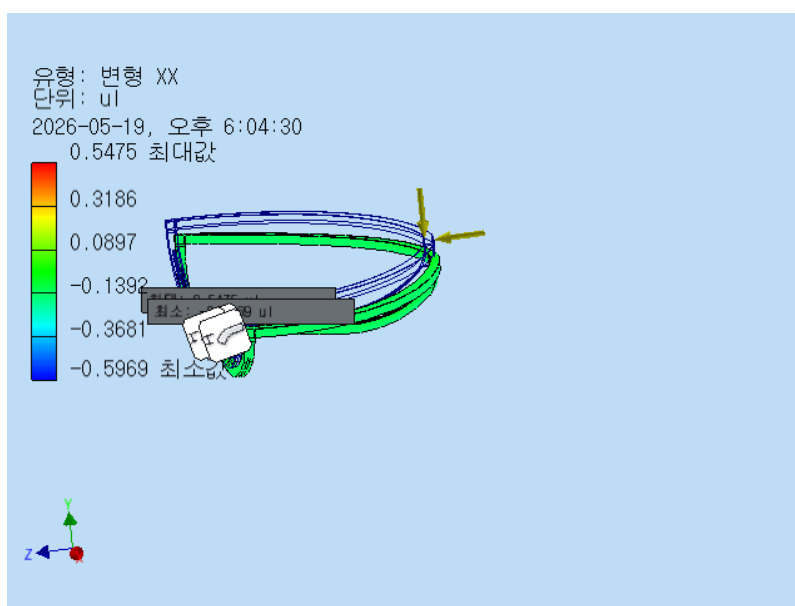
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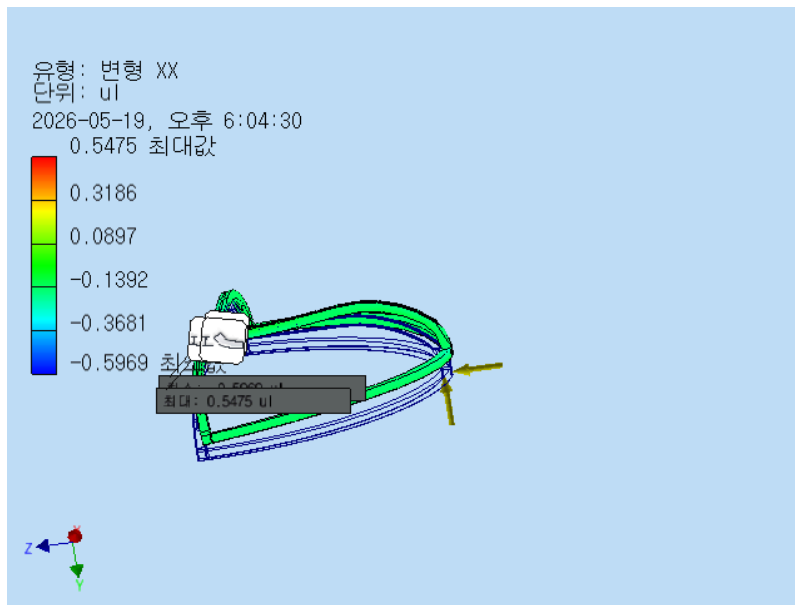


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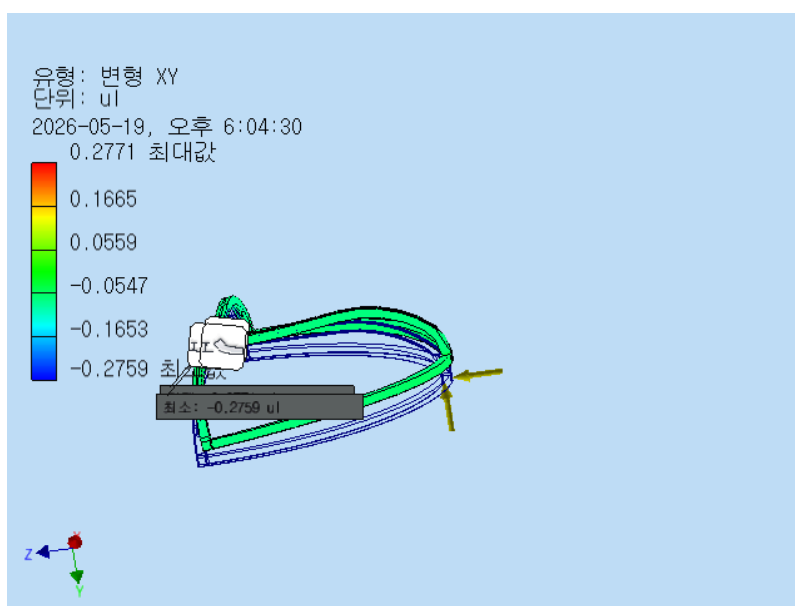
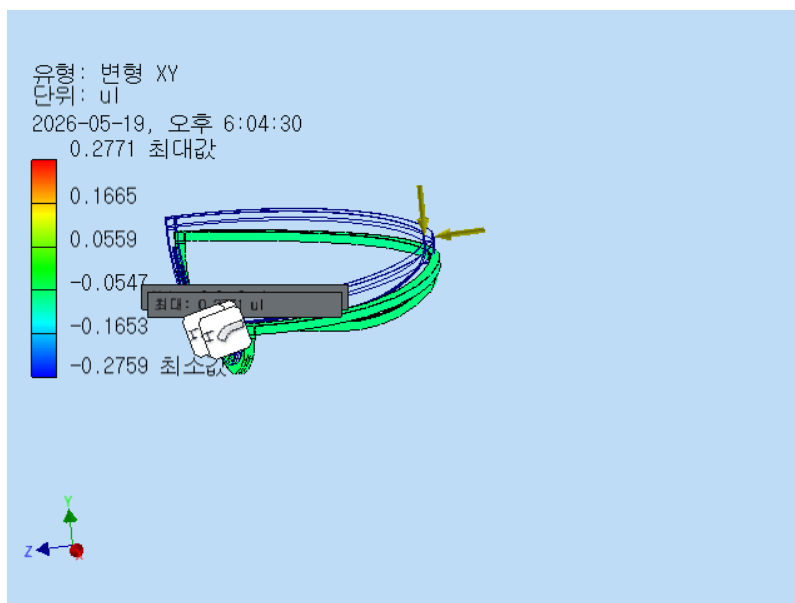


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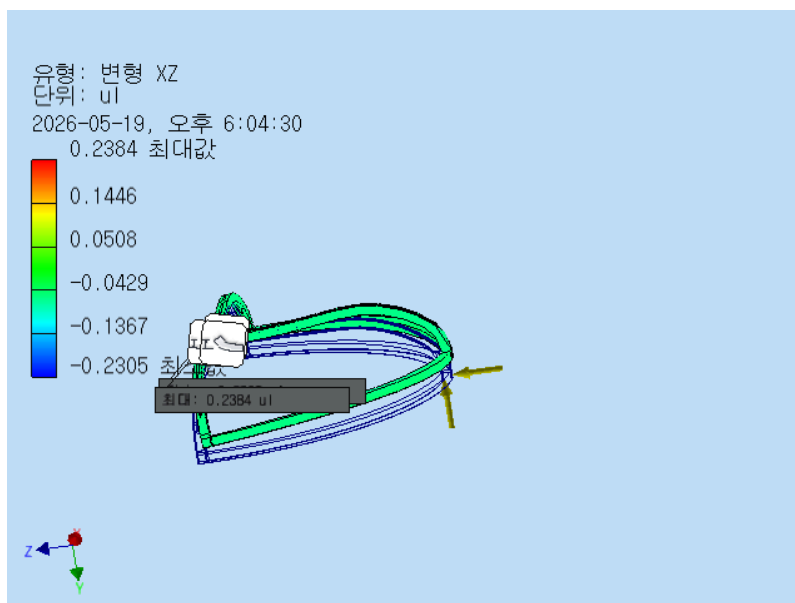
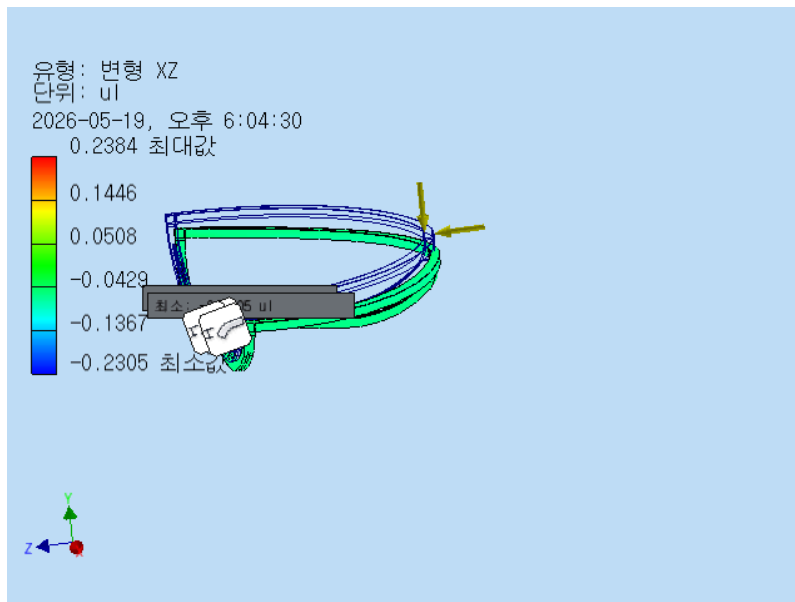




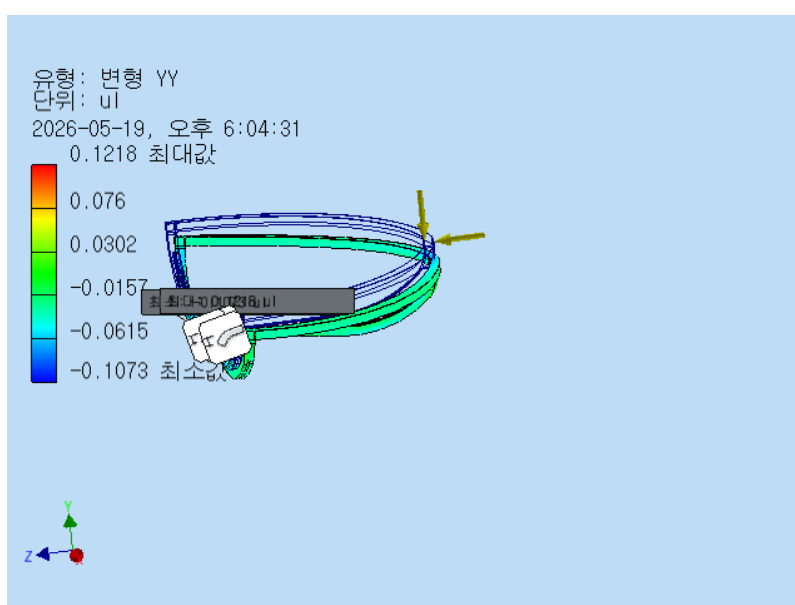
변형 XY



변형 XZ

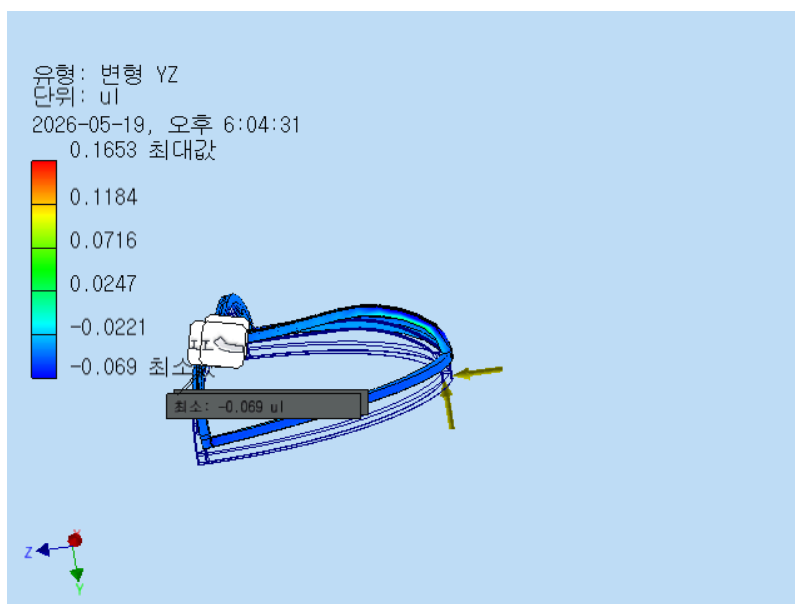
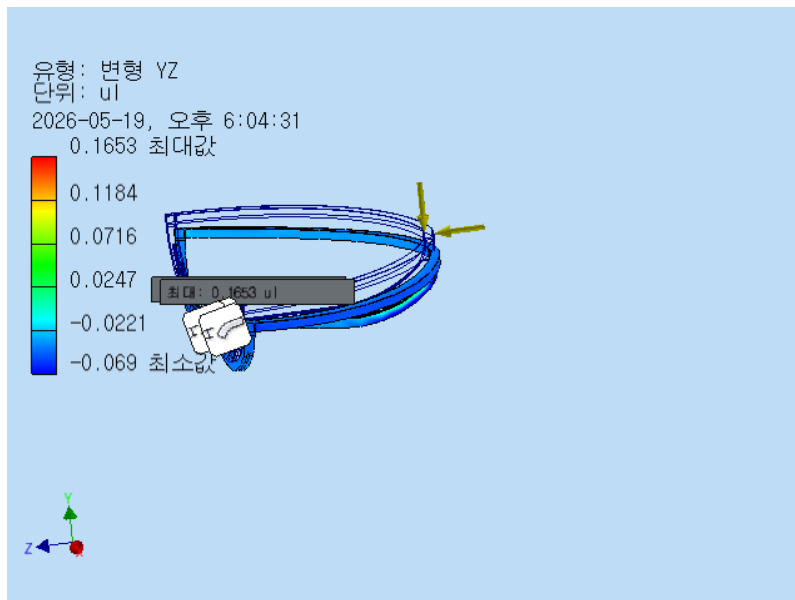


변형 YY

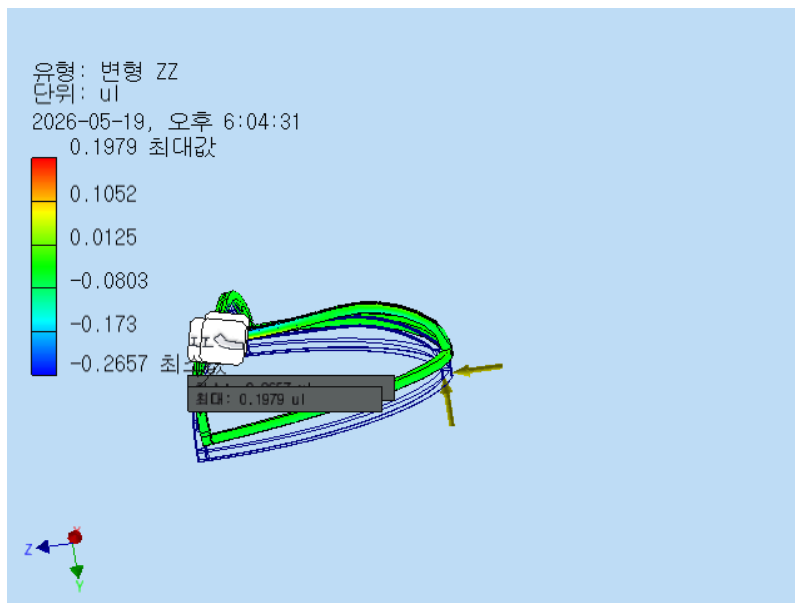
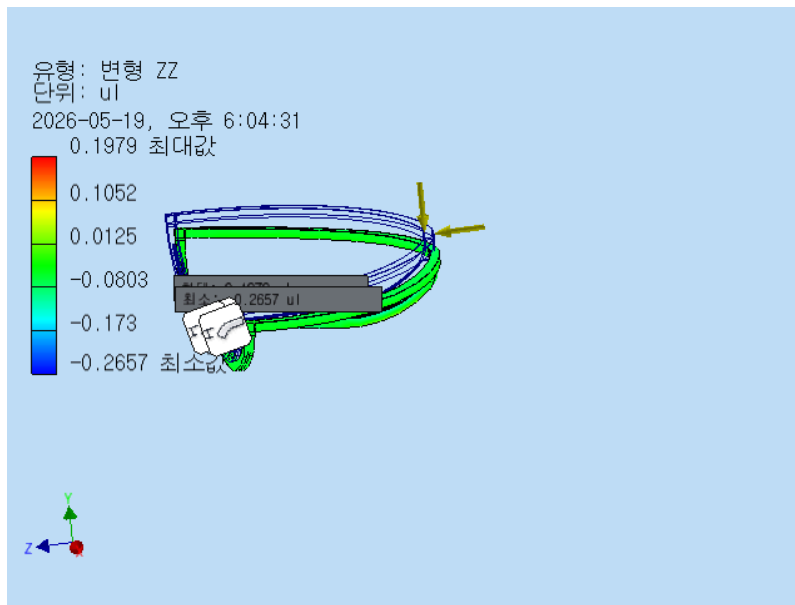




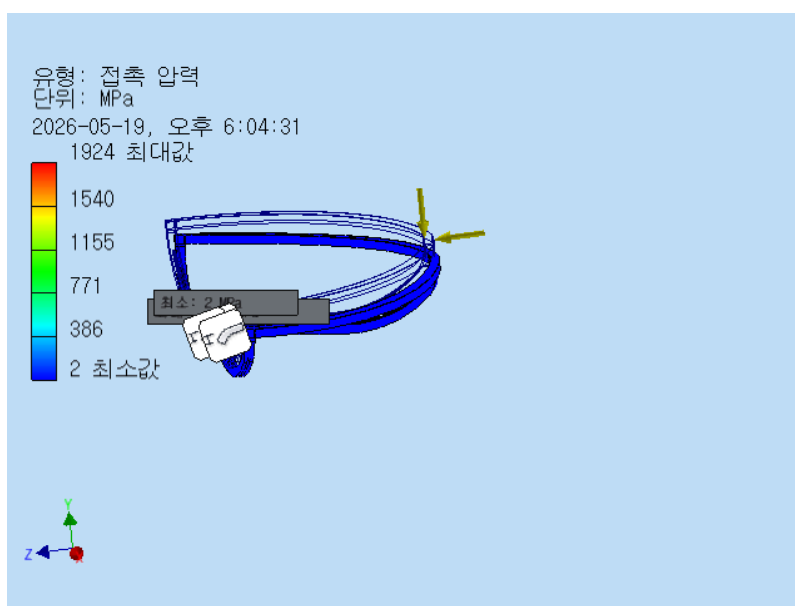
변형 YZ

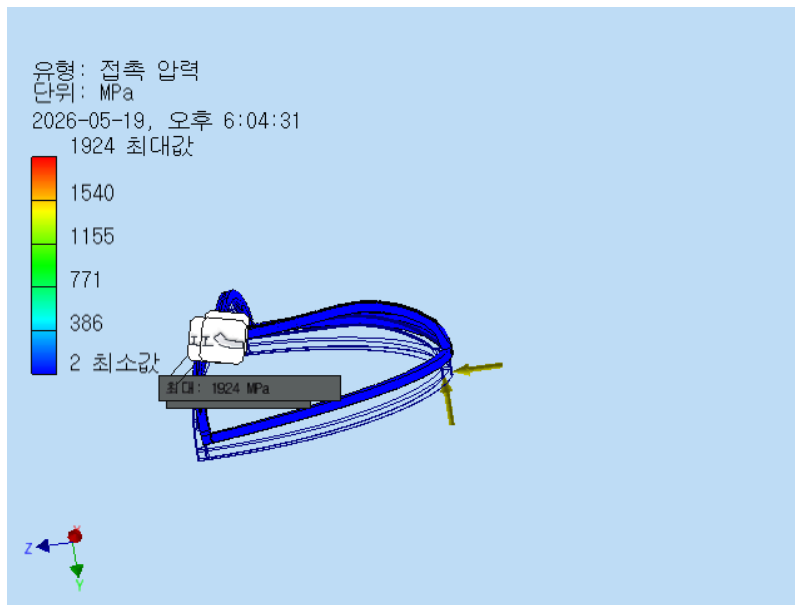


변형 ZZ

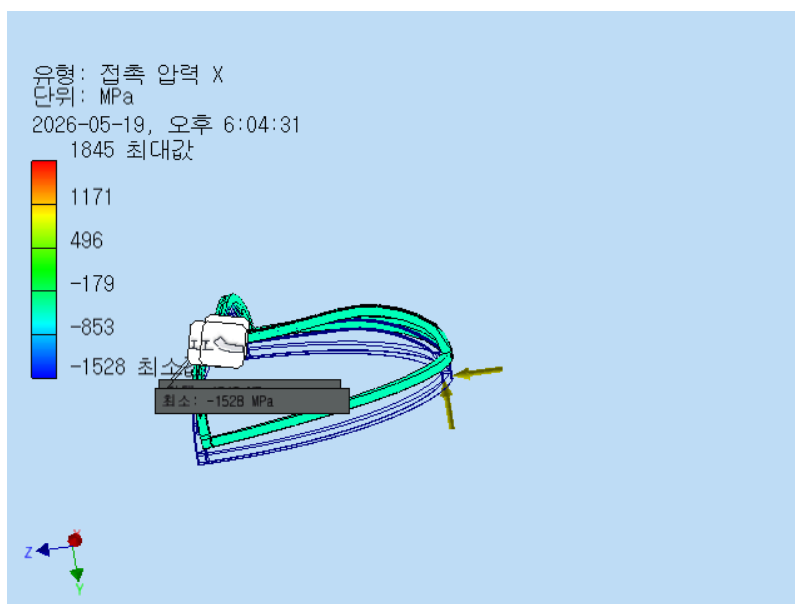
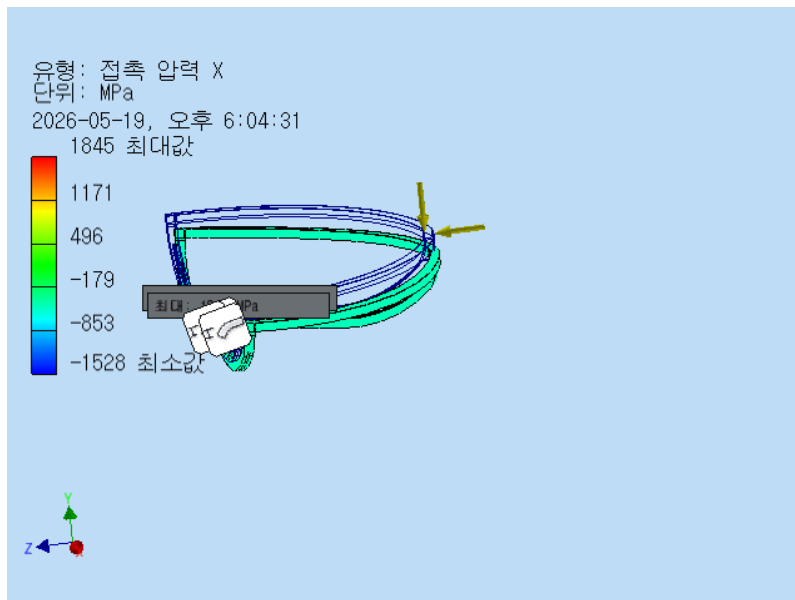


접촉 압력

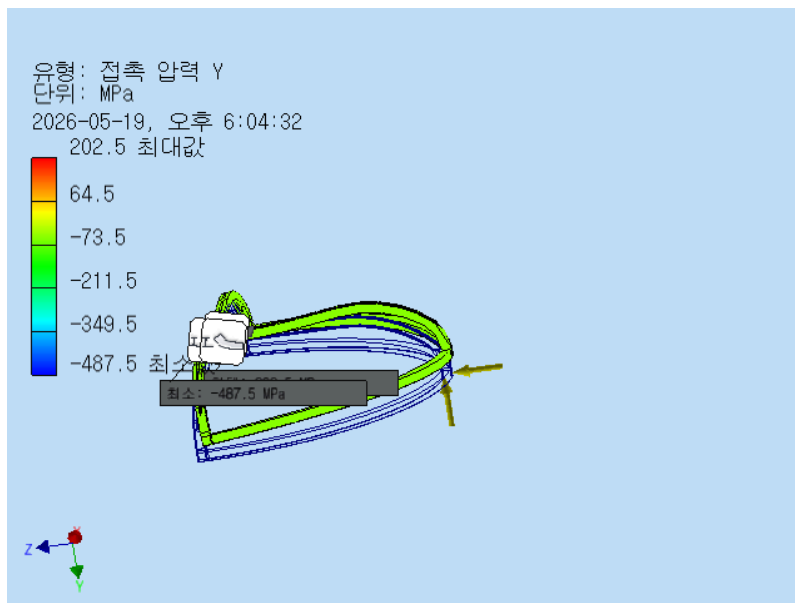
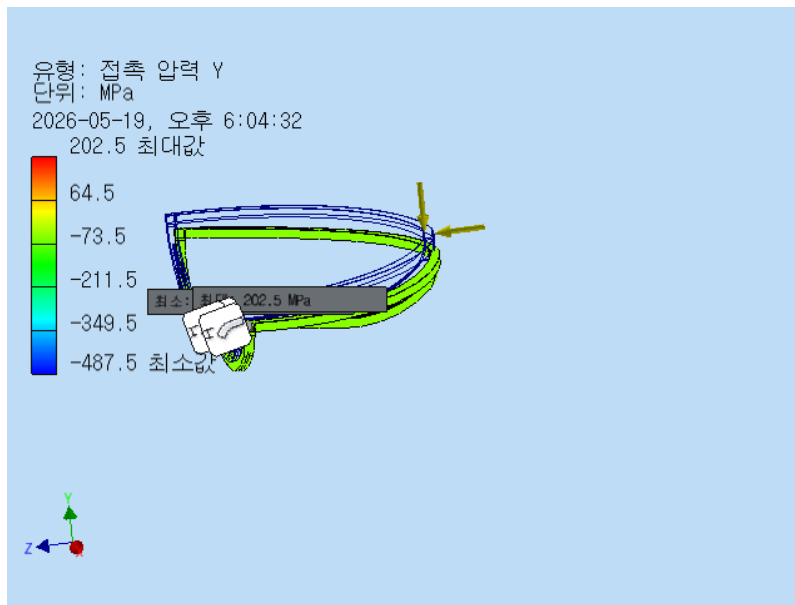




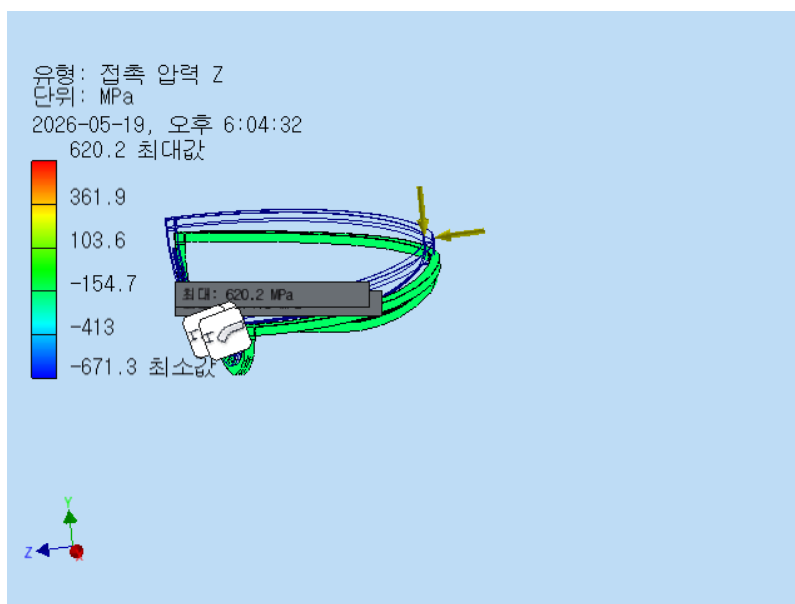
접촉 압력 X

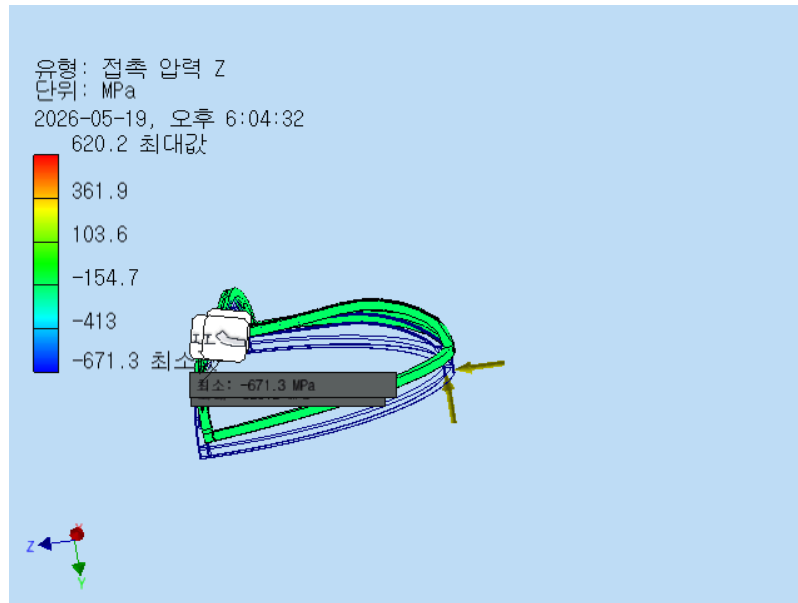


접촉 압력 Y



접촉 압력 Z





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